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A Myopia bonus and malus

Notice that the myopia gap consists of two pieces: a *myopia bonus*, the improvement that can be obtained at the current forward pass by ignoring the future forward passes; and a *myopia malus*, the cost to the future forward passes that is incurred by not pre-caching for them. To be precise, in the untied case,[9] given a choice of myopic $\tilde{\boldsymbol{\theta}}_1, \ldots, \tilde{\boldsymbol{\theta}}_n$ satisfying constraints (1) of Definition 1, write

$$\begin{aligned}
&\ell(\tilde{\boldsymbol{\theta}}_1, \ldots, \tilde{\boldsymbol{\theta}}_n) - \min_{\boldsymbol{\theta}_1,\ldots,\boldsymbol{\theta}_n} \ell(\boldsymbol{\theta}_1, \ldots, \boldsymbol{\theta}_n) \\
&= \sum_{i=1}^{n} \left(\min_{\boldsymbol{\theta}_{i+1},\ldots,\boldsymbol{\theta}_n} \ell(\tilde{\boldsymbol{\theta}}_1, \ldots, \tilde{\boldsymbol{\theta}}_{i-1}, \tilde{\boldsymbol{\theta}}_i, \boldsymbol{\theta}_{i+1}, \ldots, \boldsymbol{\theta}_n) - \min_{\boldsymbol{\theta}_i,\ldots,\boldsymbol{\theta}_n} \ell(\tilde{\boldsymbol{\theta}}_1, \ldots, \tilde{\boldsymbol{\theta}}_{i-1}, \boldsymbol{\theta}_i, \boldsymbol{\theta}_{i+1}, \ldots, \boldsymbol{\theta}_n) \right) \\
&= \sum_{i=1}^{n} \left(\ell_i(\tilde{\boldsymbol{\theta}}_1, \ldots, \tilde{\boldsymbol{\theta}}_{n-1}, \tilde{\boldsymbol{\theta}}_i) - \ell_i(\tilde{\boldsymbol{\theta}}_1, \ldots, \tilde{\boldsymbol{\theta}}_{n-1}, \boldsymbol{\theta}_i^{*i}) \right) \\
&\quad + \sum_{i=1}^{n} \sum_{j=i+1}^{n} \left(\ell_j(\tilde{\boldsymbol{\theta}}_1, \ldots, \tilde{\boldsymbol{\theta}}_{i-1}, \tilde{\boldsymbol{\theta}}_i, \boldsymbol{\theta}_{i+1}^{*i+1} \ldots, \boldsymbol{\theta}_n^{*i+1}) - \ell_j(\tilde{\boldsymbol{\theta}}_1, \ldots, \tilde{\boldsymbol{\theta}}_{i-1}, \boldsymbol{\theta}_i^{*i}, \boldsymbol{\theta}_{i+1}^{*i}, \ldots, \boldsymbol{\theta}_n^{*i}) \right).
\end{aligned}$$

where we define

$$\boldsymbol{\theta}_i^{*i}, \ldots, \boldsymbol{\theta}_n^{*i} \in \arg\min_{\boldsymbol{\theta}_i,\ldots,\boldsymbol{\theta}_n} \ell(\tilde{\boldsymbol{\theta}}_1, \ldots, \tilde{\boldsymbol{\theta}}_{i-1}, \boldsymbol{\theta}_i, \boldsymbol{\theta}_{i+1} \ldots, \boldsymbol{\theta}_n).$$

That is, the myopia gap is the sum $c + d = \sum_i c_i + \sum_i d_i$ of the myopia bonuses

$$c_i(\tilde{\boldsymbol{\theta}}_1, \ldots, \tilde{\boldsymbol{\theta}}_n) := \ell_i(\tilde{\boldsymbol{\theta}}_1, \ldots, \tilde{\boldsymbol{\theta}}_{n-1}, \tilde{\boldsymbol{\theta}}_i) - \ell_i(\tilde{\boldsymbol{\theta}}_1, \ldots, \tilde{\boldsymbol{\theta}}_{n-1}, \boldsymbol{\theta}_i^{*i}) \leq 0,$$

[9]The tied case is analogous, so we do not write it explicitly.

(the inequality following from the myopia constraints (1)), and the myopia maluses

$$\begin{aligned} & d_i(\tilde{\boldsymbol{\theta}}_1, \ldots, \tilde{\boldsymbol{\theta}}_n) \\ & \quad := \sum_{j=i+1}^{n} \left(\ell_j(\tilde{\boldsymbol{\theta}}_1, \ldots, \tilde{\boldsymbol{\theta}}_{i-1}, \tilde{\boldsymbol{\theta}}_i, \boldsymbol{\theta}_{i+1}^{*i+1} \ldots, \boldsymbol{\theta}_n^{*i+1}) - \ell_j(\tilde{\boldsymbol{\theta}}_1, \ldots, \tilde{\boldsymbol{\theta}}_{i-1}, \boldsymbol{\theta}_i^{*i}, \boldsymbol{\theta}_{i+1}^{*i}, \ldots, \boldsymbol{\theta}_n^{*i}) \right) \\ & \quad \geq 0, \end{aligned}$$

with $d_i + c_i \geq 0$ for each $1 \leq i \leq n$ by the definition of the $\boldsymbol{\theta}_j^{*i}$. *A priori*, a small myopia gap does not necessarily imply a small (in magnitude) myopia bonus c and malus d. Indeed, in the case when the myopia gap is small, a large value for c (and thus a corresponding large value for d) means precisely that the transformer model is committing significant resources to pre-caching that could otherwise have been used to improve inference on the current position. On the other hand, it is possible that both c and d are small; that is, there is not much cost associated with pre-caching for the future, as the present forward pass already results in information (breadcrumbs) useful for that purpose.

In practice, the myopia bonus may be difficult to estimate, as it depends on the result of $O(n)$ separate optimization problems (each of which, in practice, is a full transformer model training run). Thus, we instead compute the local myopia bonus of Definition 4.

A.1 Explicit gradient paths

The dependence of forward pass G_i on previous forward passes G_j for $j < i$ is mediated through hidden states $\boldsymbol{h}_1, \ldots, \boldsymbol{h}_{i-1}$:

$$G_i(\mathbf{x}_1, \ldots, \mathbf{x}_i; \boldsymbol{\theta}_1, \ldots, \boldsymbol{\theta}_i) = \tilde{G}_i(\boldsymbol{h}_1, \ldots, \boldsymbol{h}_{i-1}, \mathbf{x}_i; \boldsymbol{\theta}_i)$$

where the $\boldsymbol{h}_i$ are themselves recursively defined parameterized functions $\boldsymbol{h}_i = \boldsymbol{h}_i(\boldsymbol{h}_1, \ldots, \boldsymbol{h}_{i-1}, \mathbf{x}_i; \boldsymbol{\theta}_i)$. (Note that we are making a choice here to consider transformers as functions of hidden states, and not their key/value states. This has implications for how myopic descent is defined: when hidden state $\boldsymbol{h}_j$ is attended to by forward pass $i > j$, we consider the key and value weights $\boldsymbol{W}_K$ and $\boldsymbol{W}_V$, respectively, to belong to forward pass i. Thus, they are updated by the gradient wrt. ℓ_i.)

With the hidden states explicitly written out, the gradient wrt the loss is a sum over all possible paths to the present: for $j < i$,

$$\frac{\partial G_i}{\partial \boldsymbol{\theta}_j}(\mathbf{x}_1, \ldots, \mathbf{x}_i; \boldsymbol{\theta}_1, \ldots, \boldsymbol{\theta}_i) = \sum_{j=i_1<\ldots<i_m<i} \frac{\partial \boldsymbol{h}_j}{\partial \boldsymbol{\theta}_j} \frac{\partial \hat{G}_i}{\boldsymbol{h}_{i_m}} \prod_{k=1}^{m-1} \frac{\partial \boldsymbol{h}_{i_k}}{\partial \boldsymbol{h}_{i_{k-1}}}.$$

where the sum is over all partitions $j = i_1 < \ldots < i_m < i$.

B The myopic attention mechanism

An important primitive that we use when implementing the myopic gradient descent of Section 3.4, the local myopic bonus of Definition 4, and the transformer bigram in Section 5.1 is an attention mechanism that uses key/value states for past forward passes differing from those it uses for the current pass, while still computing all forward passes in parallel. We call our construction the *myopic attention mechanism*. We use it to implement several distinct transformer training methodologies:

- When training with *myopic descent*, the past key/value states are the result of key/value weights $\boldsymbol{W}_K$ and $\boldsymbol{W}_V$, respectively, multiplying a cloned and detached copy of the previous hidden states.
- During both training and inference of a *local myopic model*, the past key/value states come from a separate frozen pre-trained transformer model.
- During both training and inference of a *transformer bigram model*, the past key/value states are simply zeroed out.

Let $\boldsymbol{X} = (\boldsymbol{x}_1, \ldots, \boldsymbol{x}_n)^\top \in \mathbb{R}^{n \times d}$ be the sequence of residual stream hidden states per position, with each row representing one position's hidden state in $\mathbb{R}^d$. Let $\boldsymbol{W}_Q, \boldsymbol{W}_K, \boldsymbol{W}_V \in \mathbb{R}^{d \times h}$ be the query, key, value weight matrices for one attention head, of dimensionality h, in the transformer G. Denote

$$\boldsymbol{Q} := \boldsymbol{X}\boldsymbol{W}_Q,\ \boldsymbol{K} := \boldsymbol{X}\boldsymbol{W}_K,\ \boldsymbol{V} := \boldsymbol{X}\boldsymbol{W}_V$$

and let $\tilde{\boldsymbol{Q}}, \tilde{\boldsymbol{K}}, \tilde{\boldsymbol{V}}$ be the alternate states we wish to use for off-diagonal attention terms. We adopt the convention that lowercase letters with subscripts represent rows of matrices; e.g. $\boldsymbol{q}_i$ is the ith row of $\boldsymbol{Q}$. For simplicity of presentation, we omit causal masking; the modifications that should be made in the presence of a mask are straightforward.

Recall that the vanilla attention mechanism for G is

$$\boldsymbol{Y} = \sigma(\boldsymbol{Q}\boldsymbol{K}^\top)\boldsymbol{V},$$

where σ is row-wise softmax. Writing this out token-wise,

$$\boldsymbol{y}_i = Z_i^{-1} \sum_{j=1}^{n} \exp(\boldsymbol{q}_i^\top \boldsymbol{k}_j)\mathbf{v}_j,$$

where Z_i is the partition function

$$Z_i := \sum_{j=1}^{n} \exp(\boldsymbol{q}_i^\top \boldsymbol{k}_j).$$

The myopic attention mechanism, on the other hand, is written tokenwise as

$$\begin{aligned}
\tilde{\boldsymbol{y}}_i &= \tilde{Z}_i^{-1} \left(\exp(\boldsymbol{q}_i^\top \boldsymbol{k}_i)\mathbf{v}_i + \sum_{j \neq i} \exp(\boldsymbol{q}_i^\top \tilde{\boldsymbol{k}}_j)\tilde{\mathbf{v}}_j \right) \\
&= \tilde{Z}_i^{-1} \sum_{j=1}^{n} \exp(\boldsymbol{q}_i^\top \tilde{\boldsymbol{k}}_j + \delta_{ij}\boldsymbol{q}_i^\top(\boldsymbol{k}_j - \tilde{\boldsymbol{k}}_j))(\tilde{\mathbf{v}}_j + \delta_{ij}(\mathbf{v}_j - \tilde{\mathbf{v}}_j)) \\
&= \sum_{j=1}^{n} a_{ij}\tilde{\mathbf{v}}_j - \sum_{j=1}^{n} \delta_{ij}a_{ij}(\mathbf{v}_j - \tilde{\mathbf{v}}_j)
\end{aligned}$$

where

$$\begin{aligned}
\tilde{Z}_i &:= \exp(\boldsymbol{q}_i^\top \boldsymbol{k}_i) + \sum_{j \neq i} \exp(\boldsymbol{q}_i^\top \tilde{\boldsymbol{k}}_j) \\
&= \sum_{j=1}^{n} \exp(\boldsymbol{q}_i^\top \tilde{\boldsymbol{k}}_j + \delta_{ij}\boldsymbol{q}_i^\top(\boldsymbol{k}_j - \tilde{\boldsymbol{k}}_j)), \\
a_{ij} &:= \tilde{Z}_i^{-1} \exp(\boldsymbol{q}_i^\top \tilde{\boldsymbol{k}}_j + \delta_{ij}\boldsymbol{q}_i^\top(\boldsymbol{k}_j - \tilde{\boldsymbol{k}}_j)),
\end{aligned}$$

and δ_{ij} is the Kronecker delta. Now, notate

$$(\operatorname{diag} \boldsymbol{A})_{ij} := \delta_{ij}A_{ij}.$$

That is, $\operatorname{diag} \boldsymbol{A}$ is the diagonal matrix that has the same entries as $\boldsymbol{A}$ along the diagonal and is zero elsewhere. We are now able to write the myopic attention mechanism in matrix form:

$$\tilde{\boldsymbol{Y}} = \boldsymbol{A}\tilde{\boldsymbol{V}} + (\operatorname{diag} \boldsymbol{A})(\boldsymbol{V} - \tilde{\boldsymbol{V}}),$$

where

$$\boldsymbol{A} := \sigma(\boldsymbol{Q}\tilde{\boldsymbol{K}}^\top + \operatorname{diag}(\boldsymbol{Q}(\boldsymbol{K}^\top - \tilde{\boldsymbol{K}}^\top))).$$

C Proofs

We use two assumptions on the loss function to be minimized. These are standard in the first-order methods literature.

Definition 6. *A function $f\colon \mathbb{R}^n \to \mathbb{R}$ is called L-**smooth** if it is continuously differentiable with L-Lipschitz gradient. That is, for all $\boldsymbol{x}, \boldsymbol{y}$ in the domain,*

$$\|\nabla f(\boldsymbol{x}) - \nabla f(\boldsymbol{y})\|_2 \leq L\|\boldsymbol{x} - \boldsymbol{y}\|_2.$$

Definition 7. *A function $f\colon \mathbb{R}^n \to \mathbb{R}$ is called σ-**strongly convex** if, for all $\boldsymbol{x}, \boldsymbol{y}$ in the domain,*

$$f(\boldsymbol{y}) \geq f(\boldsymbol{x}) + \nabla f(\boldsymbol{x})^\top (\boldsymbol{y} - \boldsymbol{x}) + \frac{\sigma}{2}\|\boldsymbol{y} - \boldsymbol{x}\|_2^2.$$

In particular, recall that strong convexity implies the existence of a unique minimum. Hence, it makes sense to write, for example, $\boldsymbol{x}^* = \arg\max_{\boldsymbol{x}} f(\boldsymbol{x})$ without ambiguity.

C.1 Gradient descent with untied weights

Theorem 8. *Assume $\ell\colon \Theta^n \to \mathbb{R}$ is σ-strongly convex and L-smooth for some $\sigma, L > 0$. Consider ordinary gradient descent with untied weights*

$$\boldsymbol{\theta}_i^{(t)} = \boldsymbol{\theta}_i^{(t-1)} - \eta \nabla_{\boldsymbol{\theta}_i} \ell(\boldsymbol{\theta}_1^{(t-1)}, \ldots, \boldsymbol{\theta}_i^{(t-1)}) \qquad \forall i \in \{1, \ldots, n\}.$$

Then, for $\boldsymbol{\theta}_1^*, \ldots, \boldsymbol{\theta}_n^* = \arg\min_{\boldsymbol{\theta}_1, \ldots, \boldsymbol{\theta}_n} \ell(\boldsymbol{\theta}_1, \ldots, \boldsymbol{\theta}_n)$, for small enough $\eta > 0$,

$$\|\boldsymbol{\theta}_i^{(t)} - \boldsymbol{\theta}_i^*\|_2^2 \leq \left(1 - \frac{2\eta\sigma L}{\sigma + L}\right)^t \|\boldsymbol{\theta}_i^{(0)} - \boldsymbol{\theta}_i^*\|_2^2 \qquad \forall i \in \{1, \ldots, n\}.$$

Proof. This is a standard convergence result for gradient descent on strongly convex functions. For example, see Nesterov (2018). □

C.2 Gradient descent with tied weights

Theorem 9. *Assume ℓ is σ-strongly convex and L-smooth, and consider ordinary gradient descent with tied weights*

$$\begin{aligned}
&\boldsymbol{\theta}_1^{(0)} = \ldots = \boldsymbol{\theta}_n^{(0)} \\
&\boldsymbol{\theta}_i^{(t+1)} = \boldsymbol{\theta}_i^{(t)} + \eta \sum_{i=1}^n \nabla_{\boldsymbol{\theta}_i} \ell(\boldsymbol{\theta}_1, \ldots, \boldsymbol{\theta}_n) \qquad \forall i \in \{1, \ldots, n\}.
\end{aligned}$$

There exists $\eta > 0$ such that

$$\|\boldsymbol{\theta}_i^{(t)} - \boldsymbol{\theta}^*\|_2^2 \leq \left(1 - \frac{2\sqrt{n}\eta\sigma L}{\sigma + L}\right)^t \|\boldsymbol{\theta}_i^{(0)} - \boldsymbol{\theta}^*\|_2^2 \qquad \forall i \in \{1, \ldots, n\}.$$

where $\boldsymbol{\theta}^* = \arg\min_{\boldsymbol{\theta}} \ell(\boldsymbol{\theta}, \ldots, \boldsymbol{\theta})$.

Proof. This is again the standard convergence result, now applied to descent with step size $\sqrt{n}\eta$ on the σ-strongly convex L-smooth function $\boldsymbol{\theta} \mapsto \ell(\boldsymbol{\theta}/\sqrt{n}, \ldots, \boldsymbol{\theta}/\sqrt{n})$. Alternatively, one may think of this as projected gradient descent constrained to the subspace $\boldsymbol{\theta}_1 = \ldots = \boldsymbol{\theta}_n$ with a step size of $\sqrt{n}\eta$. Projected gradient descent inherits the same convergence properties as unconstrained gradient descent (Beck, 2017). □

C.3 Myopic descent with untied weights

Theorem 10. *Assume each of the $\ell_1, \ldots, \ell_n$ are σ-strongly convex and L-smooth. Consider myopic gradient descent with untied weights*

$$\boldsymbol{\theta}_i^{(t)} = \boldsymbol{\theta}_i^{(t-1)} - \eta \nabla_{\boldsymbol{\theta}_i} \ell_i(\boldsymbol{\theta}_1^{(t-1)}, \ldots, \boldsymbol{\theta}_i^{(t-1)}) \qquad \forall i \in \{1, \ldots, n\}.$$